

Constanza Inez C. de Dios, Ph.D.

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Summary

Biostatistician with background in biobehavioral research. Proficient in applied data analysis in clinical research using both frequentist and Bayesian statistical inference. Enthusiastic project collaborator dedicated to systematic problem solving.

Highlights

- 7 years of biostatistician experience in clinical research
- Data science approaches to analyzing physiological, behavioral, and clinical outcomes data
- Clinical trial and experimental research design
- Data reduction and wrangling
- Fluent in R
- Experience in REDCap management
- Experience in Python, SAS, Stata, and SPSS
- Management of multiple projects
- Effective instructor and speaker

Experience

Instructor (Non-Tenure Research)

Center for Neurobehavioral Research on Addiction
Department of Psychiatry and Behavioral Sciences

April 2023—Current

UTHealth Houston
Houston, Texas, USA

- Analyze psychiatric, physiological, EHR, genomic, biomarker, clinical, and health outcome data via generalized linear models, multilevel modeling, multivariate statistics, time series modeling, predictive modeling, and statistical learning.
 - Contexts: randomized controlled trials, observational studies, longitudinal designs, service delivery, quality improvement.
 - Content areas: substance use, mood disorders, mental health, health disparities, neurology, emergency medicine.
- Apply alternative statistical tools to non-experimental data including generalized additive models, meta-analyses, and latent variable mixture models.
- Apply frequentist or Bayesian inference as appropriate.
- Evaluate safety and efficacy of therapeutics in psychiatry and respiratory disease.
- Perform Bayesian simulations for power and trial sample size calculations.
- Identify most important predictors of substance misuse and treatment outcomes through machine learning.
- Serve as co-investigator on grants funded by federal and state agencies.
- Write statistical analysis plans, write statistical sections on grants, author publications, and address statistician reviews.
- Advise masters students, interns, and postdoctoral fellows on statistical methods.
- Present findings at research conferences.
- Inform structure of case report forms and electronic data capture.
- Process and analyze cumulative total 7.3GB (460k observations, 8k variables) across 60 projects.

Graduate Researcher and Instructor

Cognitive Electrophysiology and Clinical Neuroscience Lab
Department of Psychology

September 2013—August 2019

University of South Florida
Tampa, Florida, USA

- Applied electrophysiological methods in studying reward salience and visual attention.
- Signal-processed a cumulative 120GB of EEG sensor data from 300 human participants over five separate experiments.
- Analyzed sensor data using principal components analyses and repeated-measures analyses of variance (resulted in 9 poster presentations).
- Supervised 8 students in signal-processing 56GB electrophysiological data.
- Crafted and delivered in-person course lectures and learning assessments for 45 students every semester as lab instructor and instructor of record.

Education and Training

Postdoctoral Fellowship

Applied Biostatistics in Behavioral Science, UTHealth Houston

September 2019—March 2023

Houston, Texas, USA

Doctor of Philosophy

Psychology, University of South Florida

September 2013—August 2019

Tampa, Florida, USA

Bachelor of Science

Psychology, St. Mary's College of California

September 2008—May 2012

Moraga, California, USA